



ECEN 314 - Signals and Systems

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Module 10 - Properties of Fourier Series

Learning Objectives



- Be familiar with the various properties of Fourier Series:
 - Linearity
 - Time-Shifting
 - Frequency Shifting/Modulation
 - Conjugation, Even and Odd Signals, Re and Im parts
 - Differentiation/Integration
 - Multiplication/Convolution
- Be able to use properties of FS to simplify the process of finding the FS coefficients of various periodic signals.

Finding Fourier Series Coefficients



The exponential Fourier series is described by

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o t} \leftrightarrow X_m = \frac{1}{T_o} \int_{T_o} x(t) e^{-jm\omega_o t} dt$$

For signals that have a complicated form, finding the F.S. coefficients by evaluating the integral above can often be difficult. Rather than evaluating complicated integrals by hand, we often resort to

- ✓ Evaluate the integrals numerically using MATLAB, Python, etc.
- ✓ Use various properties of F.S. that allow us to relate the F.S. coefficients of the signal in question in terms of signals whose F.S. coefficients are known or easier to find.
- ✓ Relate the F.S. coefficients to Laplace Transforms.

In these slides, we'll look at the second (and third) approach. In the process we'll also learn how performing various operations on signals alters their frequency content.

Linearity



For two signals $x(t)$ and $y(t)$ (both assumed to have the same period) whose F.S. coefficients are X_k and Y_k , respectively, suppose we form a linear combination of these two signals

$$z(t) = ax(t) + by(t)$$

for constants a and b .

$$\begin{aligned} z(t) &= a \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_0 t} + b \sum_{k=-\infty}^{\infty} Y_k e^{jk\omega_0 t} \\ &= \sum_{k=-\infty}^{\infty} (aX_k + bY_k) e^{jk\omega_0 t} \\ Z_k &= aX_k + bY_k \end{aligned}$$

$$ax(t) + by(t) \leftrightarrow aX_k + bY_k.$$

Stated another way, the concept of superposition applies to finding Fourier series coefficients.

Time Shifting



For a positive number t_o ,

- ✓ $x(t - t_o)$ represents a time-shifted version of $x(t)$ by t_o time units to the right.
- ✓ $x(t + t_o)$ represents a time-shifted version of $x(t)$ by t_o time units to the left.

If $x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o t}$, then

$$\begin{aligned} x(t - t_o) &= \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o(t-t_o)} \\ &= \sum_{k=-\infty}^{\infty} [X_k e^{-jk\omega_o t_o}] e^{jk\omega_o t} \\ x(t - t_o) &\leftrightarrow X_k e^{-jk\omega_o t_o} \end{aligned}$$

Note that:

- ✓ $|X_k e^{-jk\omega_o t_o}| = |X_k|$ (time shifting does not change the magnitude of the FS coefficients)
- ✓ $\angle(X_k e^{-jk\omega_o t_o}) = \angle X_k - k\omega_o t_o$ (time shifting causes a phase shift in the FS coefficients)

Frequency Shifting



Suppose $x(t) \leftrightarrow X_k$ and $y(t) \leftrightarrow Y_k = X_{k-k_o} \rightarrow$ FS coefficients of $y(t)$ are frequency shifted versions of the FS coefficients of $x(t)$. Then

$$y(t) = \sum_{k=-\infty}^{\infty} Y_k e^{jk\omega_o t} = \sum_{k=-\infty}^{\infty} X_{k-k_o} e^{jk\omega_o t}$$

Making the substitution $m = k - k_o$,

$$\begin{aligned} y(t) &= \sum_{m=-\infty}^{\infty} X_m e^{j(m+k_o)\omega_o t} \\ &= e^{jk_o\omega_o t} \sum_{m=-\infty}^{\infty} X_m e^{jm\omega_o t} \\ &= x(t) e^{jk_o\omega_o t} \end{aligned}$$

Shifting the FS coefficients corresponds to multiplying by a complex exponential in the time domain.

$$x(t) e^{jk_o\omega_o t} \leftrightarrow X_{k-k_o}$$

Modulation



It is common to multiply a signal by a sine or a cosine. This process is called modulation.

$$x(t) \cos(n\omega_o t) = \frac{1}{2}x(t)e^{jn\omega_o t} + \frac{1}{2}x(t)e^{-jn\omega_o t}$$

Using the frequency shifting property from the previous slide,

$$x(t) \cos(n\omega_o t) \leftrightarrow \frac{X_{k-n}}{2} + \frac{X_{k+n}}{2}$$

Multiplying a signal in the time domain by a cosine splits the frequency content into two parts and shifts one part up and one part down in frequency.

A similar result applies to sines

$$x(t) \sin(n\omega_o t) \leftrightarrow \frac{X_{k-n}}{2j} - \frac{X_{k+n}}{2j}$$

This is how we convert a signal from one frequency band to another.

Time Reversal



$x(-t)$ is a time reversed (flipped horizontally) version of $x(t)$.

$$x(-t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o(-t)} = \sum_{k=-\infty}^{\infty} X_k e^{j(-k)\omega_o t}$$

Make the substitution $m = -k$

$$x(-t) = \sum_{m=-\infty}^{\infty} X_{-m} e^{jm\omega_o t}$$
$$x(-t) \leftrightarrow X_{-k}$$

Reversing a signal in time reverses the FS coefficients in frequency (and vice versa).

We can use time reversal to decompose a signal into even and odd parts

$$x_e(t) = \frac{x(t) + x(-t)}{2} \leftrightarrow \frac{X_k + X_{-k}}{2}$$
$$x_o(t) = \frac{x(t) - x(-t)}{2} \leftrightarrow \frac{X_k - X_{-k}}{2}$$

Conjugation, Even and Odd Signals



Recall that the FS coefficients are found from

$$X_k = \frac{1}{T_o} \int_{T_o} x(t) e^{-jk\omega_o t} dt$$

Forming a frequency reversal produces

$$X_{-k} = \frac{1}{T_o} \int_{T_o} x(t) e^{jk\omega_o t} dt$$

Consider taking the complex conjugate of the FS coefficients. If $x(t)$ is a real signal (so that $x^*(t) = x(t)$), then

$$X_k^* = \frac{1}{T_o} \int_{T_o} x(t) e^{jk\omega_o t} dt = X_{-k}$$

For even and odd parts of a signal

$$x_e(t) = \frac{x(t) + x(-t)}{2} \leftrightarrow \frac{X_k + X_{-k}}{2} = \frac{X_k + X_k^*}{2} = \text{Re}[X_k]$$
$$x_o(t) = \frac{x(t) - x(-t)}{2} \leftrightarrow \frac{X_k - X_{-k}}{2} = \frac{X_k - X_k^*}{2} = j \cdot \text{Im}[X_k]$$

Differentiation, Integration



Differentiating the FS relationship

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o t}$$

produces

$$\frac{dx}{dt} = \sum_{k=-\infty}^{\infty} jk\omega_o X_k e^{jk\omega_o t}$$
$$\frac{dx}{dt} \leftrightarrow jk\omega_o X_k$$

Applying this repeatedly generalizes the result to

$$\frac{d^n x}{dt^n} \leftrightarrow (jk\omega_o)^n X_k$$

Also,

$$\int x(t) dt \leftrightarrow \frac{X_k}{jk\omega_o} \quad \text{for } (k \neq 0)$$

Multiplication



Suppose $x(t)$ and $y(t)$ are periodic and have the same period and we form $z(t) = x(t)y(t)$.

$$\begin{aligned} z(t) = x(t)y(t) &= \left(\sum_{k=-\infty}^{\infty} X_k e^{jk\omega_o t} \right) \left(\sum_{m=-\infty}^{\infty} Y_m e^{jm\omega_o t} \right) \\ &= \sum_{k=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} X_k Y_m e^{j(m+k)\omega_o t} \end{aligned}$$

Make the substitution $n = m + k$ so that

$$\begin{aligned} z(t) &= \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} X_k Y_{n-k} e^{jn\omega_o t} \\ &= \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} X_k Y_{n-k} \right) e^{jn\omega_o t} \end{aligned}$$

Therefore,

$$z(t) = x(t)y(t) \leftrightarrow Z_n = \sum_{k=-\infty}^{\infty} X_k Y_{n-k}$$

Convolution



Suppose $x(t) \leftrightarrow X_k$ and $y(t) \leftrightarrow Y_k$. What is $z(t) \leftrightarrow Z_k = X_k Y_k$?

$$\begin{aligned} Z_k = X_k Y_k &= \left(\frac{1}{T_o} \int_{T_o} x(t) e^{-jk\omega_o t} dt \right) \left(\frac{1}{T_o} \int_{T_o} y(v) e^{-jk\omega_o v} dv \right) \\ &= \frac{1}{T_o} \int_{T_o} \frac{1}{T_o} \int_{T_o} x(t) y(v) e^{-jk\omega_o(t+v)} dt dv \end{aligned}$$

Make the substitution $\tau = t + v$ so that

$$Z_k = \frac{1}{T_o} \int_{T_o} \left[\frac{1}{T_o} \int_{T_o} x(t) y(\tau - t) dt \right] e^{-jk\omega_o \tau} d\tau$$

Therefore,

$$z(t) = \frac{1}{T_o} \int_{T_o} x(v) y(t - v) dv \leftrightarrow Z_k = X_k Y_k$$

Multiplying FS coefficients corresponds to convolution in the time domain.

Note: This is a *periodic* convolution which is defined slightly differently than the convolutions of aperiodic signals we encountered previously.

Circuit/System Response to Periodic Inputs



Consider a circuit or system that has a transfer function $H(\omega)$. Suppose we input a periodic signal $x(t) \leftrightarrow X_k$ to this circuit. We know the output signal will have FS coefficients given by

$$Y_k = X_k H(\omega_k).$$

Let $H_k = H(\omega_k)$ so that $Y_k = X_k H_k$ and define a periodic signal $\tilde{h}(t)$ which has FS coefficients, H_k .

Then, according to the convolution property on the previous slide

$$y(t) = \frac{1}{T_o} \int_{T_o} x(v) \tilde{h}(t - v) dv.$$

- ✓ We can obtain the response of a circuit/system to an arbitrary input $x(t)$ directly in the time domain by performing a periodic convolution between the input and a periodic waveform, $\tilde{h}(t)$, which is similar to, but slightly different than the impulse response of the system.
- ✓ The periodic impulse response, $\tilde{h}(t)$, would be the output the circuit/system would produce when the input has FS coefficients $X_k = 1$. $\rightarrow$ The input is a train of delta functions (comb function).

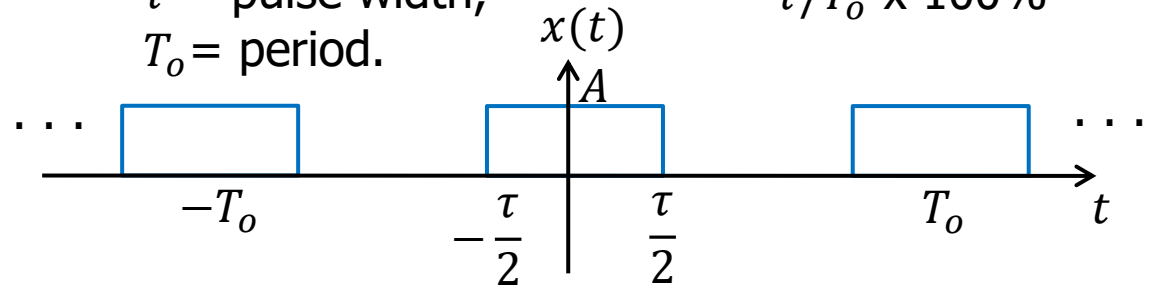
Basic Building Block – Square Wave



We will often use a generic square wave as a basic building block form which to find the FS coefficients of more complicated signals.

A = pulse height,
 τ = pulse width,
 T_o = period.

Duty cycle =
 $\tau/T_o \times 100\%$

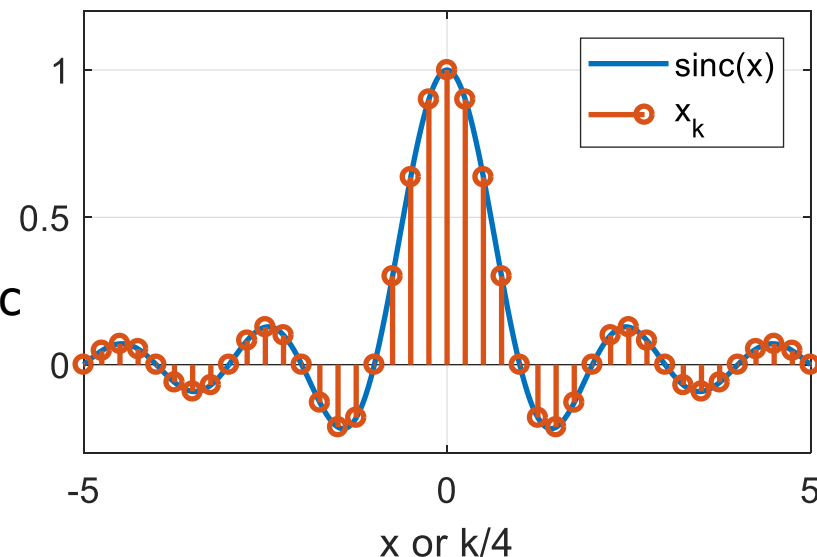


$$\begin{aligned}
 X_k &= \frac{1}{T_o} \int_{T_o} x(t) e^{-jk\omega_o t} dt = \frac{A}{T_o} \int_{-\tau/2}^{\tau/2} e^{-jk\omega_o t} dt \\
 &= -\frac{A}{jk\omega_o T_o} (e^{-jk\omega_o \tau/2} - e^{jk\omega_o \tau/2}) \\
 &= \frac{2A}{k\omega_o T_o} \sin\left(\frac{k\omega_o \tau}{2}\right) = \frac{A}{k\pi} \sin\left(\frac{k\pi\tau}{T_o}\right)
 \end{aligned}$$

It is common to write this in terms of the sinc function

$$\begin{aligned}
 \text{sinc}(x) &= \frac{\sin(\pi x)}{\pi x} \\
 X_k &= \frac{A\tau}{T_o} \text{sinc}\left(\frac{k\tau}{T_o}\right)
 \end{aligned}$$

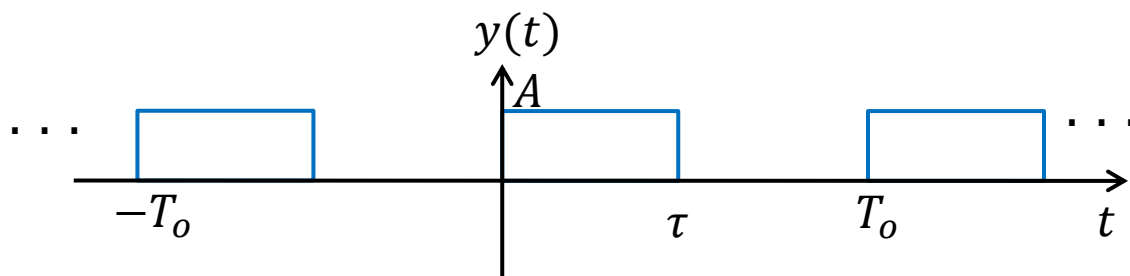
$\tau = T_o/4, A=4$



Example – Time Shifting



Example 10.1: Use the time-shifting property to find the FS coefficients of the square wave shown.

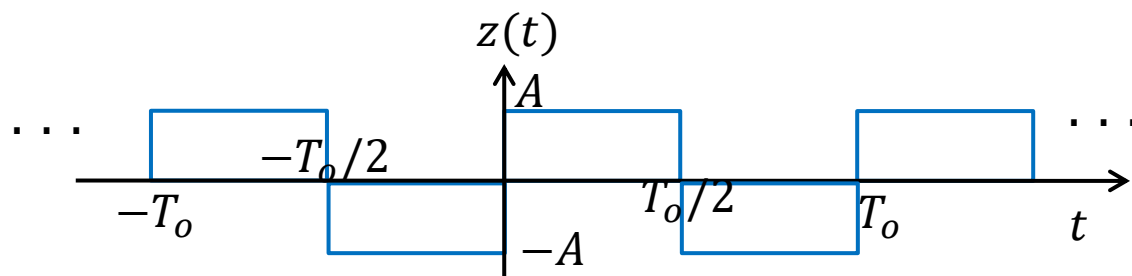


$$\text{Answer: } Y_k = \frac{A\tau}{T_o} \text{sinc}\left(\frac{k\tau}{T_o}\right) e^{-jk\omega_o\tau/2}$$

Example – Linearity



Example 10.2: Use the linearity property to find the FS coefficients of the square wave shown.



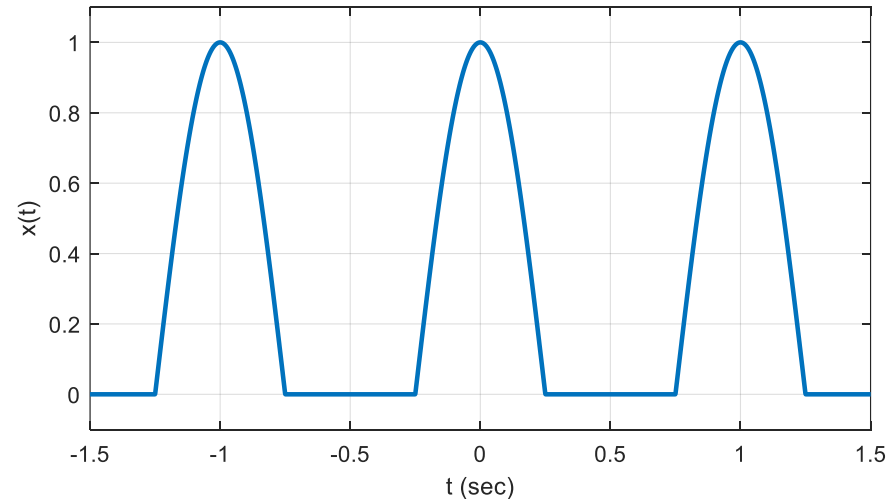
$$\text{Answer: } Z_k = \begin{cases} 0, & k \text{ even,} \\ -\frac{j2A}{k\pi}, & k \text{ odd.} \end{cases}$$

Example – Modulation



Example 10.3: Use the modulation property to find the FS coefficients of the periodic rectified sine-wave shown. One period is described by:

$$x(t) = \begin{cases} \cos(2\pi t), & |t| < 1/4, \\ 0, & \frac{1}{4} < |t| < \frac{1}{2}. \end{cases}$$

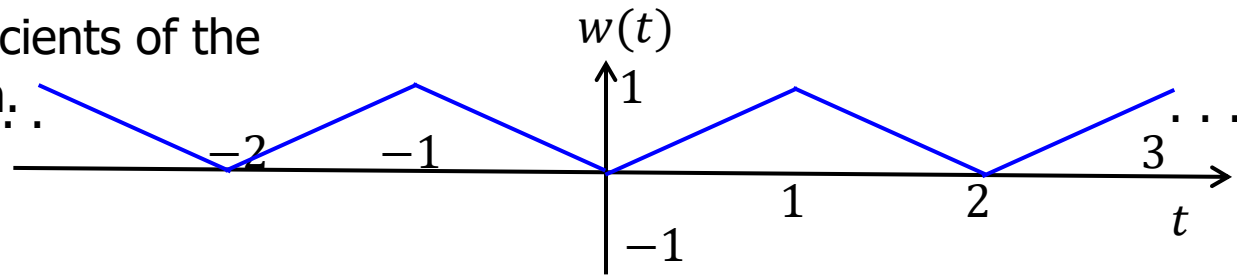


$$\text{Answer: } X_k = \begin{cases} \frac{(-1)^{k/2}}{\pi(1-k^2)}, & k \text{ even,} \\ 1/4, & k = \pm 1, \\ 0, & k \text{ odd } \neq \pm 1. \end{cases}$$

Example – Differentiation



Example 10.4: Use the differentiation property to find the FS coefficients of the periodic triangle wave shown.



$$\text{Answer: } W_k = \begin{cases} 1/2, & k = 0, \\ 0, & k \text{ even } \neq 0, \\ -\frac{2}{(k\pi)^2}, & k \text{ odd.} \end{cases}$$

Relating Fourier Series Coefficients to Laplace Transforms



Recall that the integral required to compute F.S. coefficients is

$$X_k = \frac{1}{T_o} \int_{T_o} x(t) e^{-jk\omega_o t} dt.$$

Comparing with the Laplace Transform integral, the two look similar, but not exactly the same.

$$X(s) = \int_0^{\infty} x(t) e^{-st} dt$$

Suppose we define a related signal, $\tilde{x}(t)$, as

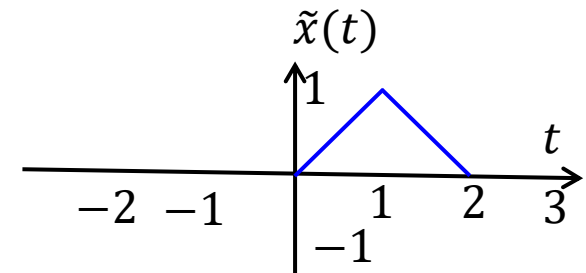
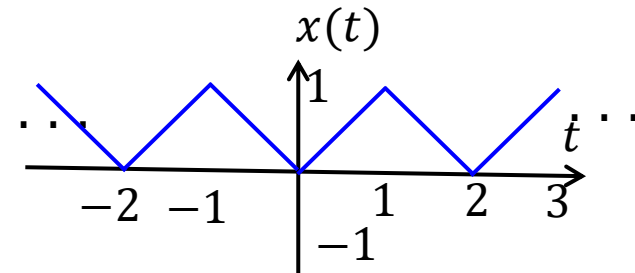
$$\tilde{x}(t) = \begin{cases} x(t), & \text{for one period,} \\ 0, & \text{otherwise.} \end{cases}$$

Then its Laplace transform is

$$\begin{aligned} \tilde{X}(s) &= \int_0^{\infty} \tilde{x}(t) e^{-st} dt \\ &= \int_{T_o} x(t) e^{-st} dt \end{aligned}$$

Then

$$X_k = \frac{1}{T_o} \tilde{X}(s) \big|_{s=jk\omega_o}.$$



$$\tilde{X}(s) = \left(\frac{1 - e^{-s}}{s} \right)^2$$

$$X_k = \frac{1}{2} \left(\frac{1 - e^{-jk\pi}}{jk\pi} \right)^2.$$